Prithvi Narayan Campus, Pokhara

Tribhuvan University

Bagar-1,Pokhara

Project Proposal

on

“RaktaSewa”

(A system connecting blood donors and seekers)

Submitted to:

Faculty of Humanities and Social Sciences

BCA program

Prithvi Naryan Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by:

Deep Darshan Thapa: 4802034

Pratik Pangeni: 4802049

Dec 2025

Abstract

RaktaSewa will be an **Android-based mobile application** designed to provide a faster, smarter, and more reliable way to connect blood donors with recipients. In Nepal, the absence of a centralized, real-time donor–recipient matching system often results in delays that can be life-threatening. This project aims to address this problem by developing a mobile platform that integrates **Firebase** for user authentication, real-time data storage, and instant communication between donors and seekers.

The system will be using three core algorithms to enhance matching efficiency. The **Donor Matching Algorithm** filters donors based on blood group compatibility, location proximity, and availability status. The **Emergency Priority Algorithm** ensures urgent requests receive immediate attention by prioritizing the nearest and most eligible donors. Additionally the **Donor Availability Prediction Algorithm** evaluates the donor's eligibility based on last donation date and recommended donation intervals. The project aims to reduce search time, improve donor engagement, and contribute to a more organized and effective blood donation network.

Keywords: RaktaSewa, Android-based mobile application, Firebase, Donor Matching Algorithm, Emergency Priority Algorithm, Donor Availability Prediction Algorithm

Contents

List of figures

1. Introduction

1.1 Background of the project

Blood plays a critical role in saving lives during medical emergencies, major surgeries, accidents, childbirth complications, and chronic illnesses. However, in Nepal, accessing blood at the right time remains a major challenge. Several regions across the country frequently face blood shortages, and families are often required to arrange blood on their own due to limited stock in hospitals and blood banks. For instance, hospitals in Banke and Nepalgunj have repeatedly reported acute shortages where relatives must find blood independently when the blood bank cannot meet demand [1]. Similar situations have been observed in Jhapa, where the Red Cross noted that blood supply levels often drop to a critically low point, lasting only a few days [2].

These challenges highlight the lack of a centralized, updated, and easily accessible donor database in Nepal. Most blood search efforts rely on personal networks, manual phone calls, and social media posts which consume valuable time during emergencies. Reports also show issues such as irregular donor participation, outdated donor information and poor accessibility in rural areas, making emergency response even more difficult [3].

To address this gap RaktaSewa is proposed as an Android-based solution that integrates real-time data management and intelligent algorithms to support faster, more organized, and reliable blood search in Nepal.

1.2 Problem Statement

Nepal continues to face recurring challenges in ensuring timely access to blood during emergencies. Many hospitals and blood banks operate with limited and inconsistent stock levels, compelling families to search for donors on their own when supplies run out. Reports from multiple districts such as Banke, Nepalgunj and Jhapa indicate frequent shortages that hinder emergency [2] [1]. Additionally, there is no unified and continuously updated digital donor database, and most donor information remains scattered, outdated, or dependent on personal contacts and social media postings [4]

Therefore, there is a need for a digital solution that can intelligently match donors with recipients, provide updated availability status, and support rapid communication during emergencies.

1.2 Objectives

- To design a mobile application that allows users to register as donors or seekers.
- To implement Firebase for real-time donor information storage and authentication.
- To develop a donor matching algorithm based on blood group compatibility, location, and availability.
- To reduce the time taken to find compatible donors during critical situation

1.4 Scope

By providing a centralized, intelligent, and real-time donor search platform, RaktaSewa helps reduce the delays caused by manual searching methods. The proposed system aims to strengthen Nepal's emergency health infrastructure through digital innovation.

1.5 Limitation

- Relies on internet connection
- Location accuracy may vary depending on device GPS and network condition
- Medical eligibility verification is based on user input

2. Literature Review

Blood donation and management systems have been a focus of research and development worldwide due to their critical role in healthcare and emergency response. In Nepal, the Nepal Red Cross Society manages blood inventory and donor information primarily through manual processes and offline records. While this system provides essential services, it often faces challenges in ensuring timely availability of blood during emergencies, particularly in remote regions where access to updated donor information is limited. Reports from districts like Banke, Nepalgunj, and Jhapa indicate frequent shortages, compelling families to rely on personal networks or social media to find donors. This demonstrates a clear gap in accessibility and real-time communication in existing systems [1] [2].

Studies on blood donation systems in Nepal and other developing countries highlight several recurring challenges. These include lack of a centralized and continuously updated donor database, delays in communication between donors and seekers, limited access in rural areas

RaktaSewa is proposed to address these gaps by providing an android based platform that integrates a centralized Firebase database for real-time donor information storage and retrieval. The system uses algorithms to match donors and seekers based on blood group compatibility, location, and availability reducing the time taken to find compatible donors

3. Methodlogy

The proposed system will be designed using a client-server architecture where the android app acts as the client and Firebase provides real-time database, authentication, and notifications. The database stores user profiles, blood groups, locations, and availability status. A donor matching algorithm considers blood group compatibility, location, and availability to connect seekers with suitable donors efficiently. And the project will follow prototyping method for system development for our specific project requirement.

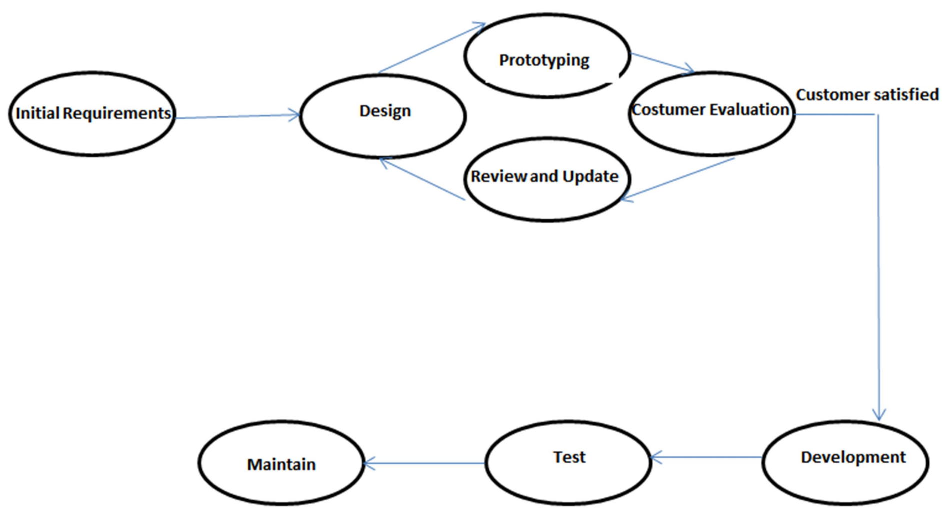


Figure 1 Prototyping model

4. System Analysis

To ensure the success of RaktaSewa, a thorough system analysis is essential. This involves understanding the challenges in blood donation management in Nepal and designing a well-defined solution. A robust analysis model helps in framing the problem and developing an effective, user-friendly system that connects donors and seekers in real time.

4.1 Requirement Identification

During this phase, we identify what the system should do and how it should work.

4.1.1 Functional Requirements

- Users should be able to register as donor or seeker
- Donors should be able to update their availability and last donation date
- Seekers should be able to find donors by blood group
- System should match donors with seekers

4.1.2 Non-Functional Requirements

- App should work on android
- User data must be stored safely
- App should be easy to use

4.2 Feasibility Study

A feasibility study has been conducted to assess the viability of the RaktaSewa, covering technical, economic, and legal aspects.

4.2.1 Technical feasibility

After the technical needs have been carefully analyzed, we have the technical skills and resources needed to system. The user interface of the app is clear and simple to use. It is made sure the website works technically

4.2.2 Economic Feasibility

No extra cost is needed for a development team. Minimal expenses are expected, covering documentation printing and website hosting.

4.2.3 Legal Feasibility:

Research has been conducted on legal requirements for operating a RaktaSewa. Steps have been taken to ensure compliance with data privacy laws and regulations.

4.3 Tools

Front-end: Java for building the app interface.

Back-end: Firebase for login, database, and notifications.

Database: Firebase Realtime Database for storing donor and seeker information.

Extra Tools: Google Maps API for location based donor matching.

4.4 System Design

The app will be designed to be simple and easy to use. Donors can update their status and location and seekers can search for donors, and the system connects them automatically.

Use case

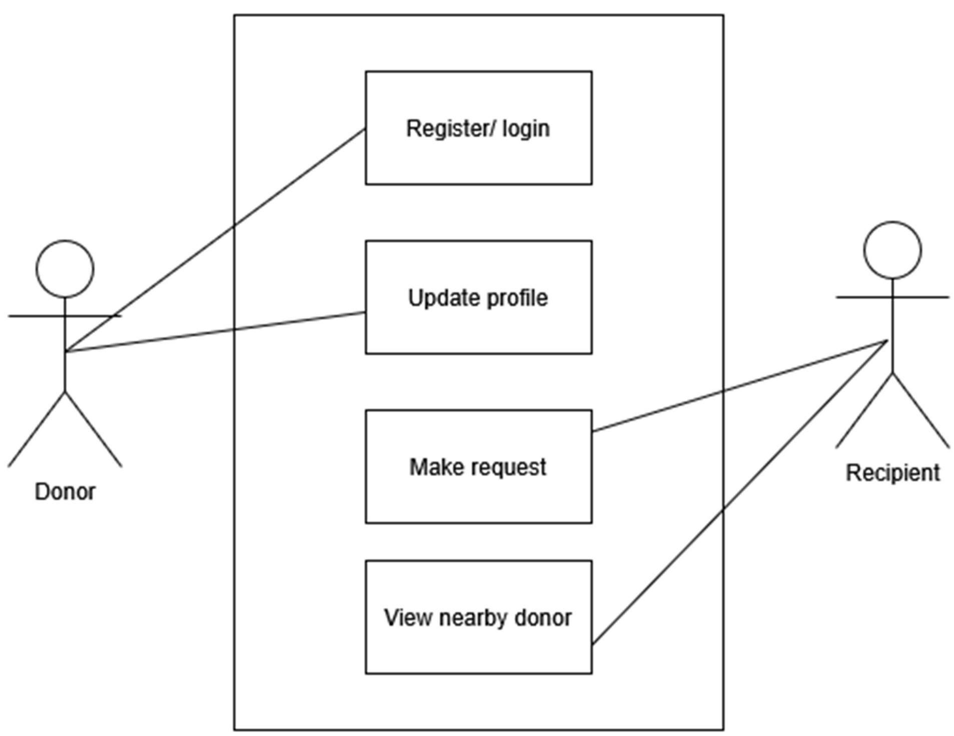


Figure 2 Use case diagram of RaktaSewa

Flow Chart

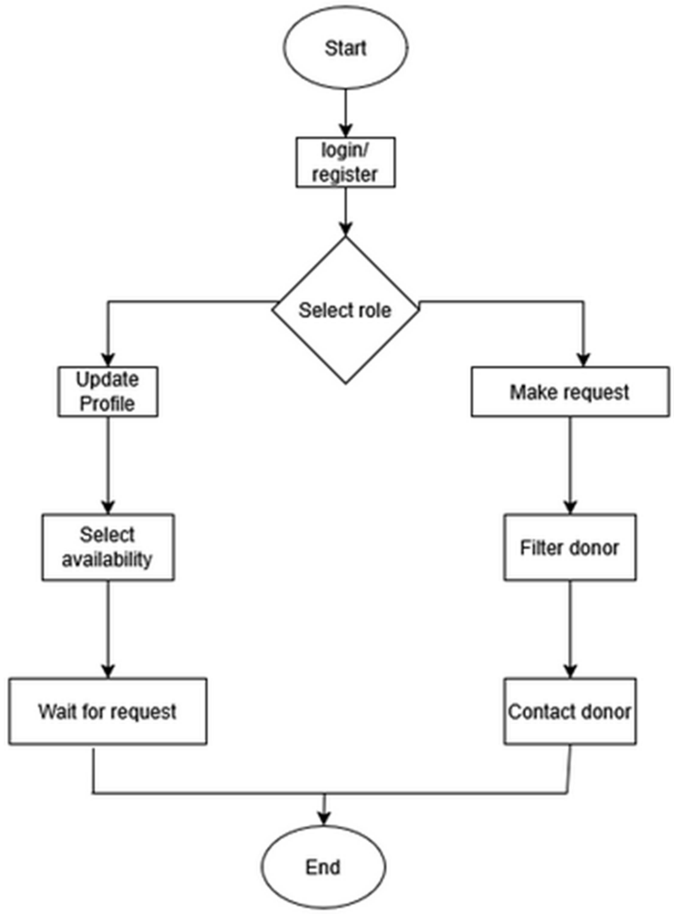
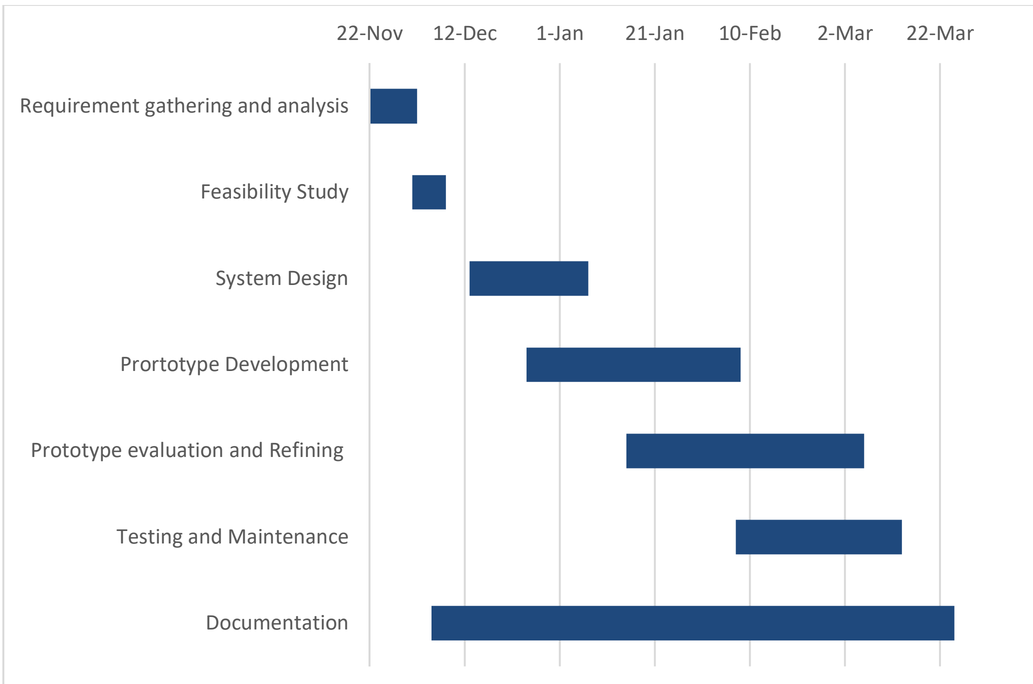


Figure 3 System flow chart

5. Gantt Chart

The Gantt chart below shows the timeline for the development of the project. It provides a visual representation of the project's key tasks their respective durations. This timeline helps to ensure that all phases of the project, from planning and design to development, testing, and deployment, are completed on schedule.



6. Expected Outcome

We will develop a mobile application that connects blood donors and seekers in a fast, reliable, and organized way. The app will allow users to register, verify their details, and participate as either donors or blood requesters. Core features such as creating blood requests, viewing available donors, sending notifications, and updating donor status will function smoothly across all devices. The system will help users find suitable donors without delays, especially during emergencies, making the process more efficient than traditional manual communication.

The application will include a smart matching system that filters donors based on blood group, location, and availability. This will reduce the time spent searching for a suitable donor and increase the chances of timely support. Users will be able to track their previous requests, donor activities, and response records, helping them better understand the overall donation process. A clean and simple interface will ensure that users with minimum technical knowledge can navigate the app easily.

In conclusion, this project aims to deliver a dependable, user-friendly, and socially impactful blood donation application that enhances the overall emergency response system. By using the Prototype Model, the system will improve through regular feedback and iterative refinement, ensuring that the final product meets the expectations of real users and contributes meaningfully to saving lives in critical situations.

7. Reference

[1] S. Khan, "Banke faces blood shortage with only 150 pints left in stock," Rising Nepal Daily, 15 June 2024. [Online]. Available: https://risingnepaldaily.com/news/44447.

[2] "Jhapa Red Cross appeals for blood donation amid acute shortage," Kharhub.com, 27 October 2025. [Online]. Available: https://english.khabarhub.com/2025/27/503083/.

[3] circleheart, "Challenges in Blood Donation in Nepal," 26 March 2025. [Online]. Available: https://circleheart.org/blog/blood-donation-and-its-importance-for-the-nepalese-community/.

[4] "Where's the blood?," Nagarik Network, 23 November 2012. [Online]. Available: https://myrepublica.nagariknetwork.com/news/53901/.